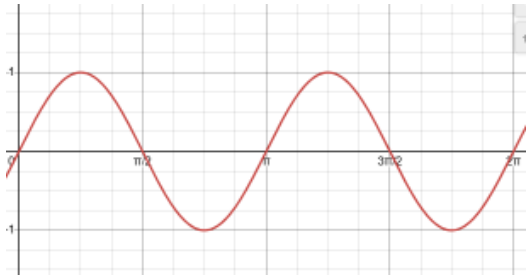
| Question | | | Answer | Marks | AO | Guidance |
|---|---|---|---|---|---|---|
| 4 | (a) | |  | **M1** | **1.1** | sine wave with 2 complete cycles between 0 and $2\pi$ which must be clear from labelling; condone eg waves of unequal amplitude or different length; allow labelling to override poor sketching; allow labelling in degrees for **M1** only |
| | | | | **A1** | **1.1** | all correct with amplitude 1; ignore graph outside $[0, 2\pi]$ |
| | | | | **[2]** | | |
| 4 | (b) | | either $-\frac{\pi}{12}$ or $-\frac{5\pi}{12}$ **oe** seen | **M1** | **3.1a** | **NB** from use of $\sin^{-1}\left(-\frac{1}{2}\right)$<br>**NB** – 0.262 or – 1.309; decimals correct to 2 dp or better |
| | | | **or** | | | **or** |
| | | | either $\frac{7\pi}{6}$ or $\frac{11\pi}{6}$ **oe** seen | | | 3.665 or 5.760; decimals correct to 2 dp or better<br>**M1** may be implied by two correct answers in radians |
| | | | $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$ | **A1** | **1.1** | two correct values<br>**NB** awrt 1.833, 2.880, 4.974, 6.021 or correct to 2 dp |
| | | | | **A1** | **1.1** | all four values correct with no extras for $0 \leq \theta \leq 2\pi$<br>if **M0A0A0** allow **SCB1** for **two** correct answers in degrees<br>**or SCB2** for all four correct in degrees 105, 165, 285 and 345 |
| | | | | **[3]** | | |